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# Rethinking CPU Performance for the Agentic AI Era

OCP 2026



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Oct 2026

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# Five Use-case Archetypes Define Today's Agentic AI Landscape

**The use case determines where the agent runs, which tools it invokes and the CPU signature it creates**

## 1 Software Engineering

WHERE IT RUNS

**Cloud developer infrastructure**

DOMINANT TOOLS

Code editing, build and test execution

DISTINCTIVE CPU SIGNATURE

**Compilation and sandboxed tests at high churn**

## 2 Enterprise Automation

WHERE IT RUNS

**Enterprise datacenter or SaaS**

DOMINANT TOOLS

APIs, records and transactions

DISTINCTIVE CPU SIGNATURE

**Latency-bounded transactions at high concurrency**

## 3 Scientific Research

WHERE IT RUNS

**HPC-adjacent facilities**

DOMINANT TOOLS

Simulation, domain tools and literature

DISTINCTIVE CPU SIGNATURE

**Numerical kernels driven by an agent loop**

## 4 Personal Assistants

WHERE IT RUNS

**Client and edge devices**

DOMINANT TOOLS

Browsing, documents, media and local models

DISTINCTIVE CPU SIGNATURE

**Client inference plus interactive parsing and rendering**

## 5 Robotics Agents

WHERE IT RUNS

**Embedded systems and fleet backends**

DOMINANT TOOLS

Perception and physical actuation

DISTINCTIVE CPU SIGNATURE

**Hard real-time control loops and middleware**

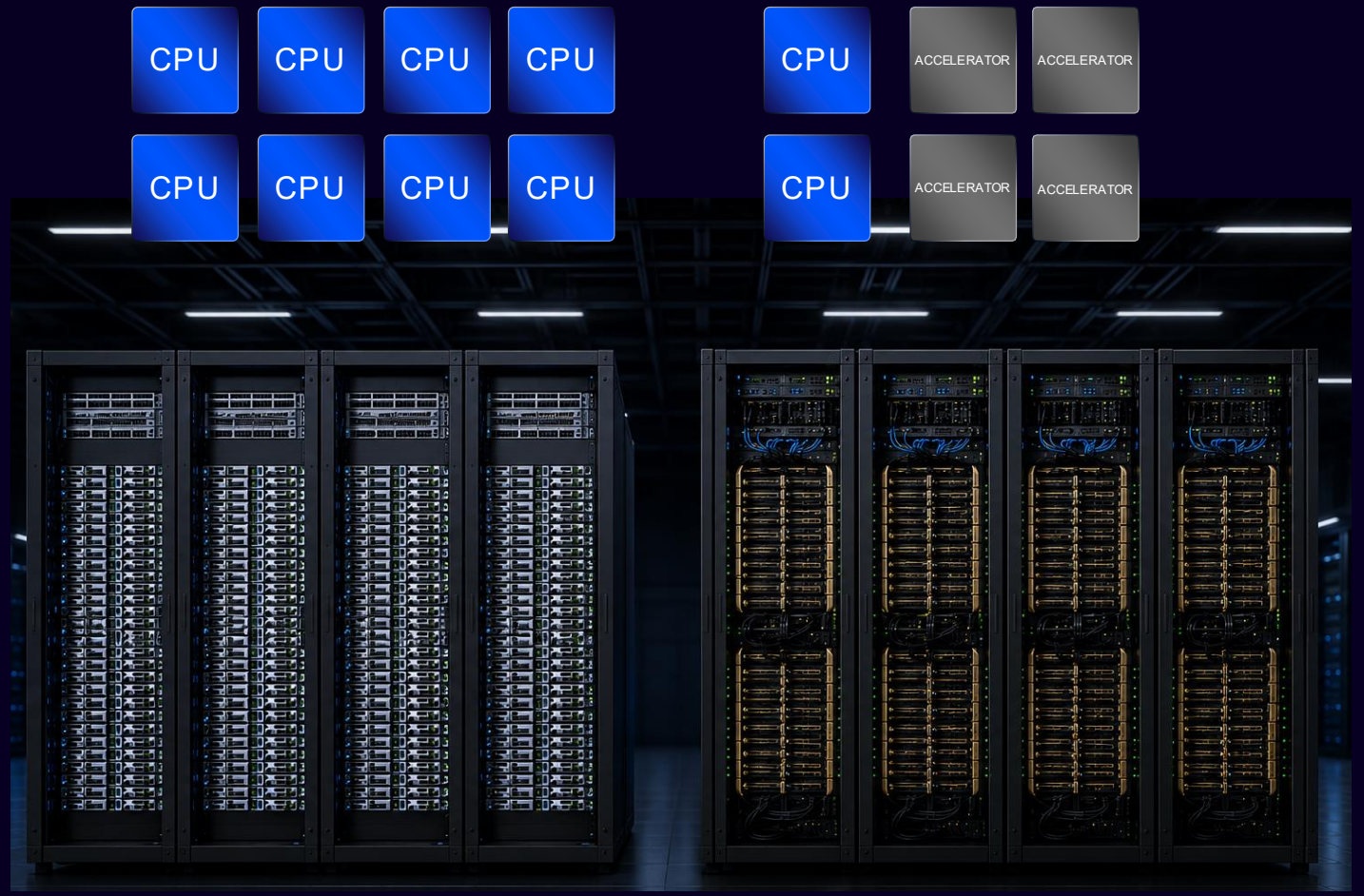
# Designing Infrastructure for Agentic AI

## Agentic Tooling & Orchestration

High-Density Compute servers

Optimized to run Multiple agents @ Scale

Balanced Throughput and Latency






Pipeline of execution spread across AI Racks and CPU Racks

Deployment Flexibility for multiple use-cases

## Agentic Reasoning & Planning

High-Performance AI Servers/Rack

Optimized for High Token throughput @ scale

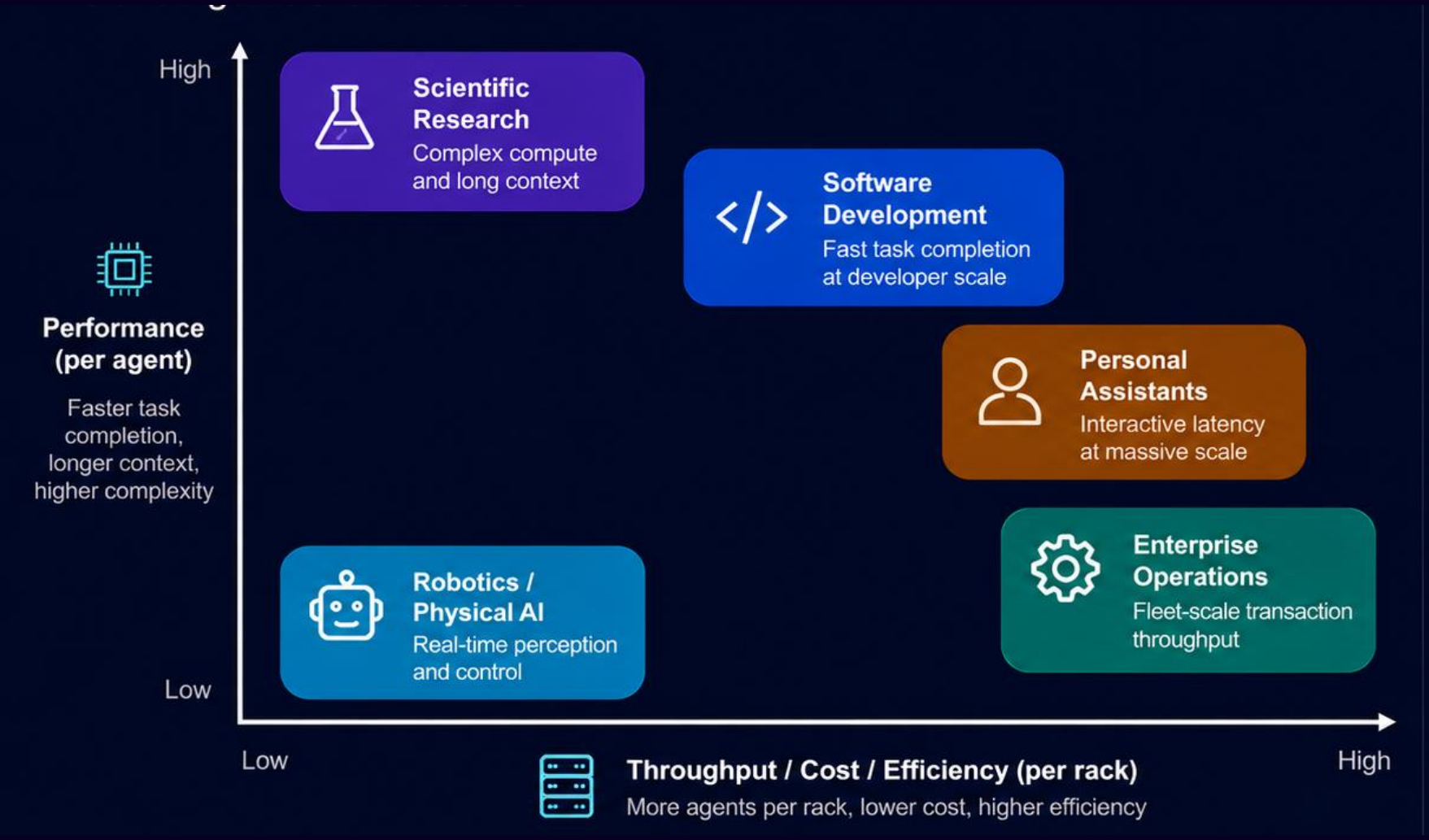
High Memory and Storage for Context



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# Use-case Requirements for Agentic Infrastructure

No two Agents are the same




**Scientific Research**
Complex analysis/simulation, long context and advanced reasoning; fewer concurrent agents.

**Software Development**
Long tool/code/test loops; high per-agent performance and developer-scale deployment.

**Personal Assistants**
Interactive response matters, with massive user scale driving high infrastructure efficiency.

**Enterprise Operations**
Large populations of independent agents executing APIs, retrieval and business workflows.

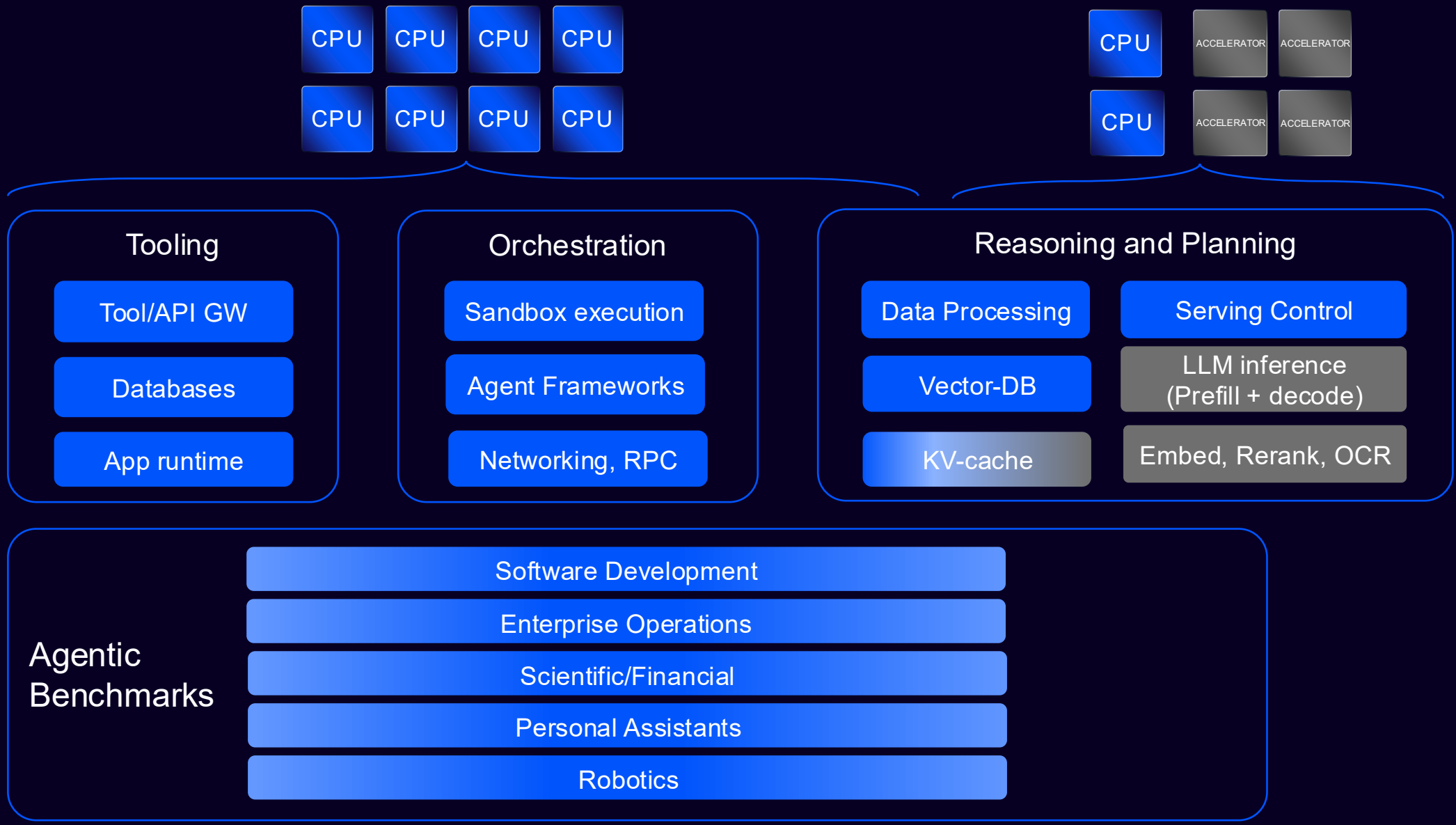
**Robotics / Physical AI**
Real-time perception/control; primarily an edge/accelerator problem rather than rack CPU density.

**Industry basis:** OpenAI Codex usage study (2026); Microsoft Discovery/BHP (2026); Salesforce Agentforce & ServiceNow EnterpriseOps-Gym (2026); Google Gemini usage study (2026); Figure Helix 02 (2026).

Relative workload characterization synthesized from published application architectures, usage data and production deployments; not direct CPU benchmark results.


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# A Benchmarking Framework for Agentic AI






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# Mapping use-cases to real-world applications and benchmark proxies

| Use case | Real-world application | Benchmark proxy |
|---|---|---|
| **Software Engineering** | Claude Code (Anthropic)<br>Codex (OpenAI)<br>Github Copilot (MSFT) | Terminal-Bench (Anthropic, Google)<br>SWE-bench (OpenAI) |
| **Enterprise Operations / Customer Service** | Agentforce (Salesforce)<br>ServiceNow AI agents<br>Sierra Agent Studio | CRMArena (Salesforce)<br>WorkArena (ServiceNow)<br>tau-bench (Sierra) |
| **Scientific/Financial Research** | DeepResearch (OpenAI)<br>Discovery (Microsoft)<br>SpectrumIQ (JPMorgan) | ScienceAgentBench<br>DiscoveryWorld<br>FinanceBenchmark (Microsoft) |
| **Personal Assistant / Content Creation** | Muse, Vibes (Meta)<br>DreamBig (BD)<br>Gemini, Copilot | GAIA2 (Meta)<br>AndroidWorld (Google)<br>AssistantBench |
| **Robotics / Physical AI** | Figure Helix | EmbodiedBench<br>Behavior1K |

***Industry-backed benchmarks provide credible proxies for the CPU demands of real-world agentic applications.***

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# From Model Benchmarks to Reproducible CPU Workloads

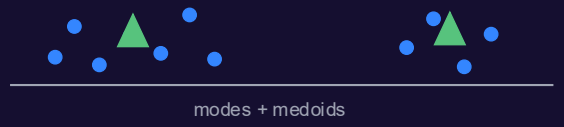
### Observe Real Agent Runs

Run an existing benchmark repeatedly. Capture the tools calls, timing and host work it generates.



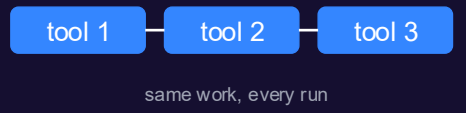

### Freeze Representative Behavior

Agent paths vary run to run. Cluster behavior and select real representative trajectories.




### Replay as CPU Workloads

Remove live model variability. Replay identical host work across systems.




**OPEN, EXTENSIBLE WORKLOAD FORMAT**

**STATE** **TOOLS** **TIMING** **CHECKS**

An open methodology: start with industry benchmarks, contribute your own trajectories, compare infrastructure consistently.

Benchmarks built to evaluate agents can become benchmarks for the infrastructure that runs them

# Learnings from running frozen trajectories across benchmarks

### Terminal-Bench

**Per-core performance matters.**

Single-agent and lightly loaded workloads are sensitive to how fast each core completes tool work.

### SWE-bench / Multi-SWE-bench

**Concurrency changes the bottleneck.**

As more agents run together, CPU, memory and storage pressure determine how many stay within SLA.

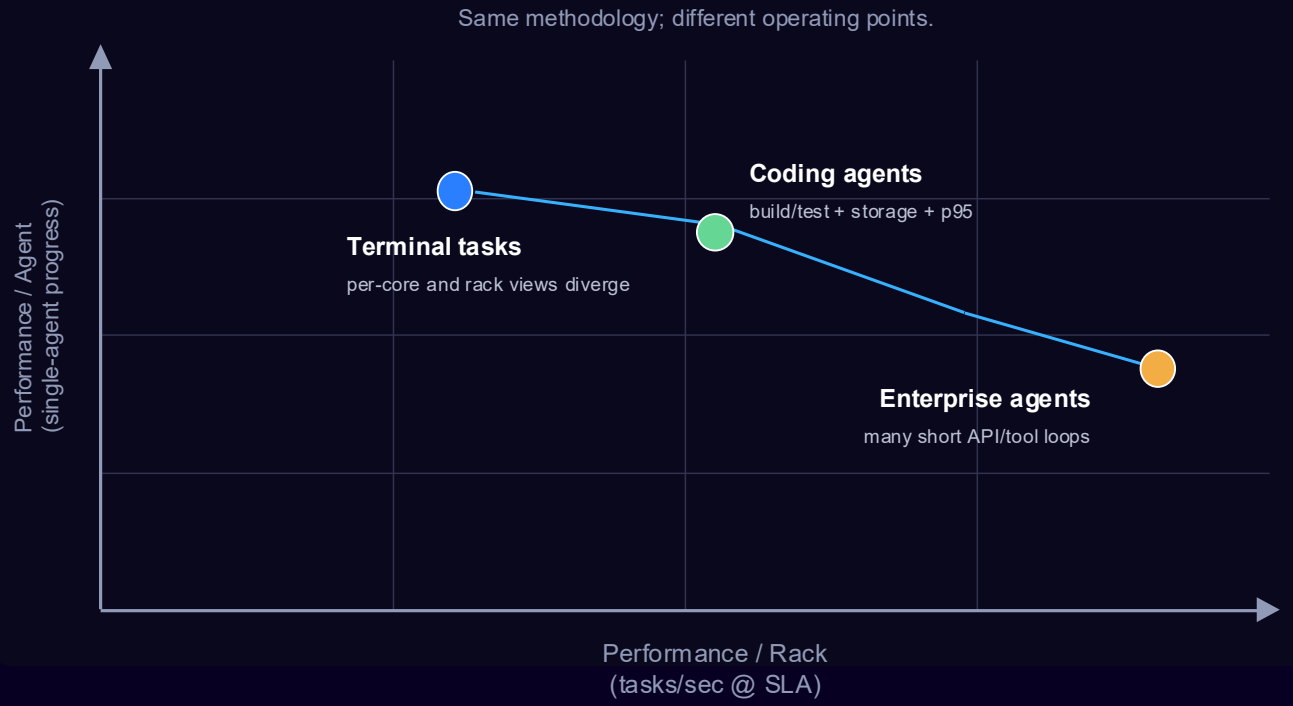
### tau-bench / tau2-bench

**Enterprise agents are a throughput problem.**

Multiple parallel, short tool/API actions repeat across many simultaneous workflows.

### The CPU question changes by workload

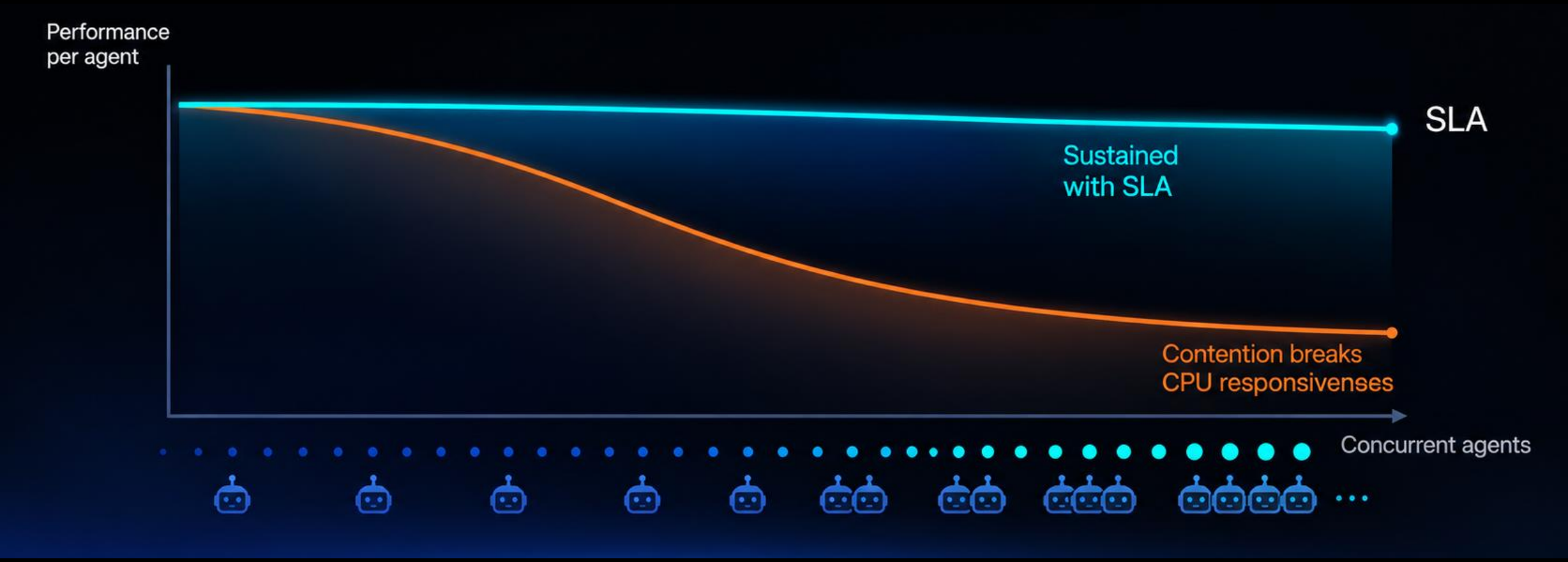
One benchmark number hides the tradeoff between single-agent progress and fleet-scale capacity.




Agentic CPU performance has two dimensions: finish each task quickly and sustain many tasks at once.

# Sustained Performance

## Concurrent agents at scale delivering task completeness





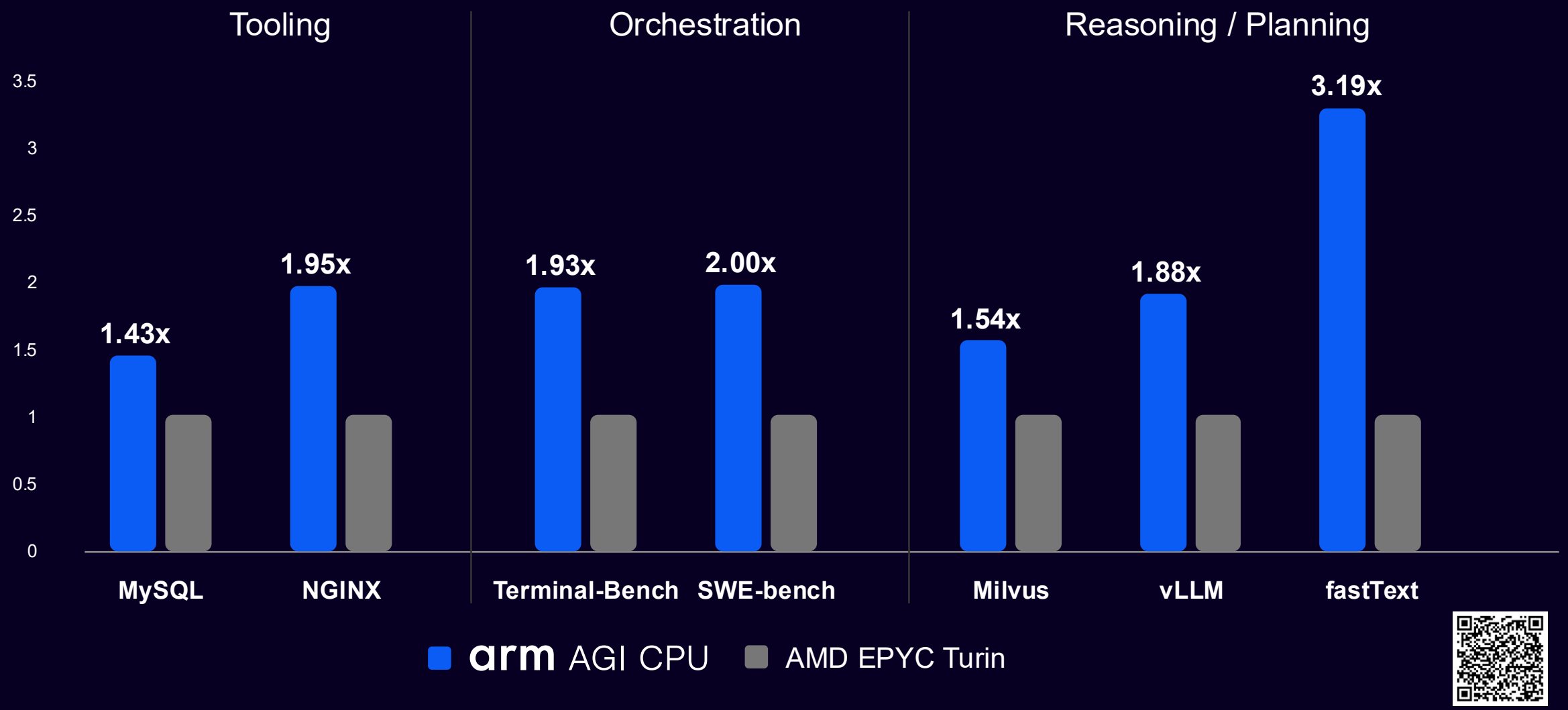
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# AGI CPU – designed for Agentic AI

| FEATURE | DESIGN | IMPACT |
| --- | --- | --- |
| 3rd Gen Neoverse cores | Micro-architecture optimized for key workloads | Optimized performance – Tooling, Orchestration, Context workloads |
| Memory latency | Co-located Compute and Memory on same die | Faster context retrieval and storage |
| Memory BW/core | 2x Higher Memory Bandwidth per core | No contention for multiple parallel complex Agentic tasks |
| Scalability | Balanced Compute, Memory and IO | Higher utilization – more Agentic tasks completed @ SLA |
| Efficiency | 200-400W allows for 1U 2N air-cooled or 1U 8N liquid cooled designs | Up to 2x Higher Rack density, Perf/Rack |

# Performance/Rack – Agentic Workloads

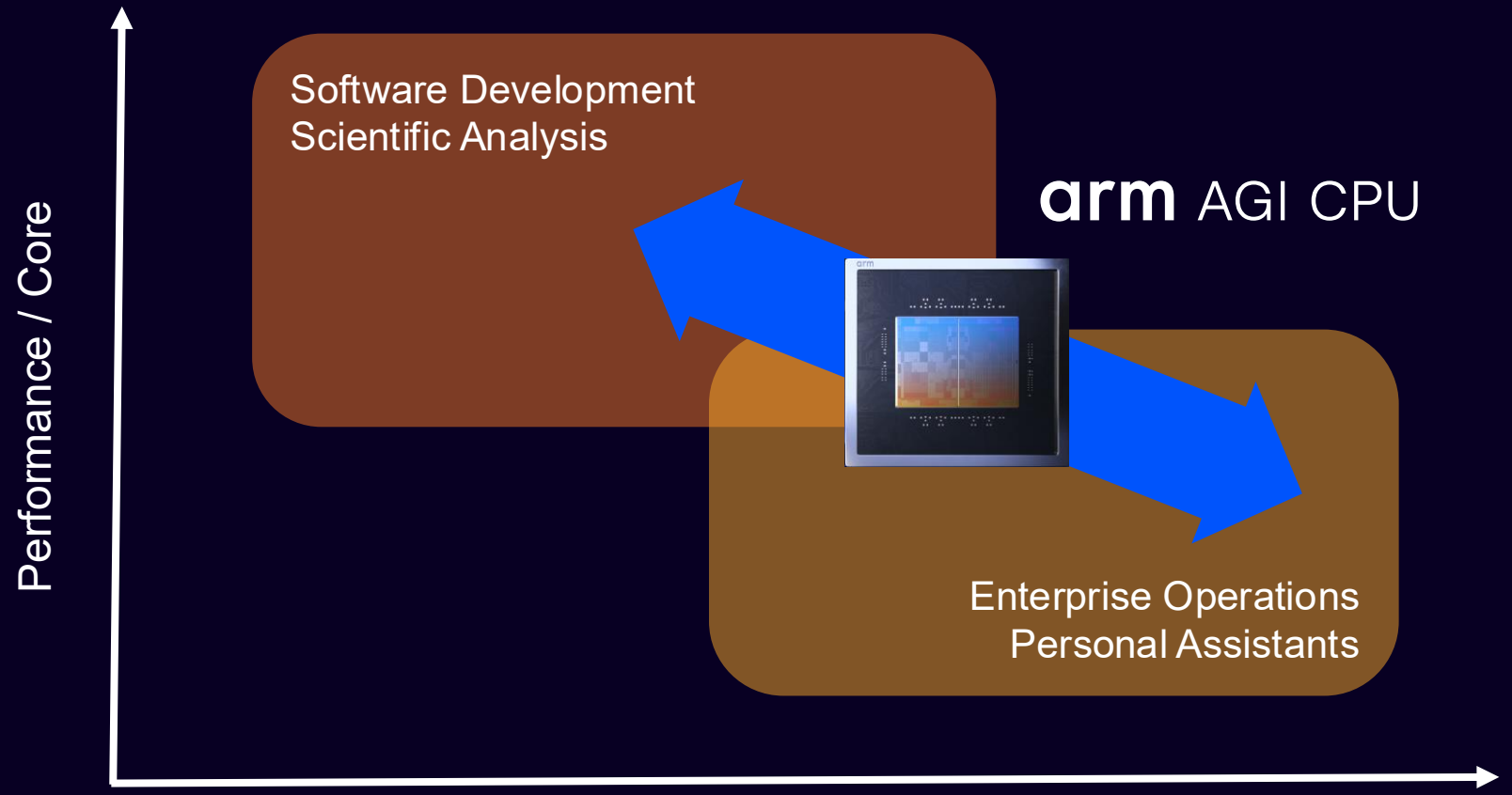
Arm AGI CPU Outperforms AMD EPYC across key Agentic tasks




Results are based on 3rd-party evaluations. All tests are conducted by Signal65 and measured on Arm AGI CPU and AMD EPYC systems and specifications noted at the time of testing. Full methodology here https://signal65.com/research/ai/arm-agi-cpu-performance-rack-scale/. Results may vary.

# Agentic AI Requires More Than One CPU Design Point

- AGI CPU combines strong per-core performance with high rack-level throughput.
- AGI CPU provides a configurable operating range to meet different use-case needs.





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Merci
Danke
Gracias
Grazie
谢谢
ありがとう
Asante
Thank you
감사합니다
धन्यवाद
Kiitos
شكرًا
ধন্যবাদ
תודה
ధన్యవాదములు
Köszönöm